About the exam

- The exam will be written, and it is 2 hours.
- It will have three questions. The first two questions have three compulsory parts; in the third, you can choose 4 out of five.
- The exam is 50% of the module, and there is no floor mark to average for assignments.
- The dates are published on the official exam website in Grangegorman.

Typical Exam Questions

Introduction to Data Mining:

1. Define Data Mining and Machine Learning (ML).

   Data Mining: The process of discovering useful patterns, trends, and relationships in large datasets using statistical and computational techniques.

   Machine Learning (ML): A branch of artificial intelligence that enables systems to learn from data and improve performance on tasks without being explicitly programmed.

2. Explore the main advantages of ML and the concept of Big Data.

   Main Advantages of Machine Learning (ML)

1. **Automation & Efficiency**
   ML automates repetitive tasks, reducing human effort and errors while accelerating processes.
2. **Predictive Insights**
   By analyzing historical data, ML models forecast trends or risk.
3. **Personalization**
   Powers recommendation systems (like Netflix) by tailoring content/products to individual user preferences.
4. **Handling Complexity**
   Excels at solving intricate problems (e.g., image recognition, natural language processing) that are impractical for manual coding.
5. **Scalability**
   Adapts to growing data volumes without significant redesign, making it ideal for dynamic environments.
6. **Continuous Improvement**
   Models self-optimize over time as they ingest new data, enhancing accuracy without explicit reprogramming.

Concept of Big Data

Definition:
Big Data refers to extremely large, diverse datasets that exceed traditional processing capabilities. It's characterized by the **3 Vs** (expanded to **5 Vs**):

1. **Volume**:
   Massive scale (e.g., terabytes to petabytes) from sources like social media, IoT devices, or transactions.
   velocity:
   High-speed generation/streaming (e.g., real-time sensor data, live financial trades).
   Variety:
   Diverse formats (structured databases, unstructured text/images, semi-structured JSON/XML).
   Veracity:
   Focus on data quality/reliability (managing noise, inconsistencies, or biases).
   Value:
   Extracting actionable insights from raw data to drive decisions.
   Why Big Data Matters:

- Enables ML/AI by providing the vast training data needed for accurate models.
- Reveals hidden patterns (e.g., customer behavior).
- Supports real-time analytics (e.g., fraud detection, emergency response).

Synergy: ML + Big Data
Big Data fuels ML by supplying rich datasets, while ML transforms this data into predictive insights—creating a cycle of innovation across industries like finance, healthcare, and logistics.

3. Enumerate applications of Data Mining.

Applications of Data Mining

1. Business & Retail

- **Market Basket Analysis** – Identifying product affinities (e.g., "customers who buy X also buy Y").
- **Customer Segmentation** – Grouping customers by behavior for targeted marketing.
- **Fraud Detection** – Detecting unusual patterns in transactions or insurance claims.
- **Churn Prediction** – Identifying customers likely to leave for retention campaigns.
- **Demand Forecasting** – Predicting sales trends for inventory management.

2. Science & Engineering

- **Climate & Environmental Studies** – Modeling climate patterns and predicting extreme events.

3. Web & Social Media

- **Recommendation Systems** – Personalized content (e.g., Netflix, Amazon, Spotify).
- **Sentiment Analysis** – Gauging public opinion from reviews or social posts.

4. Other Domains

- **Healthcare** – Predicting disease outbreaks, patient diagnosis support.
- **Finance** – Credit scoring, algorithmic trading, risk assessment.
- **Security** – Network intrusion detection, cybersecurity threat analysis.

4. Distinguish between Supervised and Unsupervised Learning, covering classification and regression.

Supervised vs. Unsupervised Learning
Here's a clear distinction between the two, with emphasis on **classification** and **regression** as key subtypes of supervised learning.

1. Supervised Learning
Core Idea: The model learns from **labeled data** (input-output pairs). Each training example includes input features and the corresponding correct output (label). The goal is to learn a mapping from inputs to outputs so that it can predict outputs for new, unseen data.
Key Characteristics:
Uses labeled datasets.
Has a clear target variable to predict.
Performance is evaluated by comparing predictions to known labels (e.g., accuracy, RMSE).
Common applications: spam detection, price prediction, image recognition.
Two Main Types:
a) Classification
Goal: Predict a **discrete class label** (category).
Output: Categorical (e.g., "spam"/"not spam", "cat"/"dog"/"bird").
Examples:

- Email spam detection (binary classification).
- Handwritten digit recognition (multi-class classification).
- Medical diagnosis (disease present/absent).

Algorithms: Logistic Regression, Decision Trees, Random Forest, SVM, Neural Networks.

b) Regression
Goal: Predict a **continuous numerical value.**
Output: A real number (e.g., price, temperature, age).
Examples:

- Predicting house prices based on features.
- Forecasting sales revenue.
- Estimating temperature trends.

Algorithms: Linear Regression, Polynomial Regression, Ridge/Lasso Regression, Regression Trees.
Summary for Supervised Learning:
Input (Features) + Known Output (Label) → Model → Predict Output for New Data

2. Unsupervised Learning
Core Idea: The model learns from **unlabeled data** (only input features, no output labels). The goal is to discover hidden patterns, structures, or groupings within the data.
Key Characteristics:
Uses unlabeled datasets.
No target variable; the system tries to learn the underlying distribution.
Harder to evaluate since there's no ground truth.
Common applications: customer segmentation, anomaly detection, recommendation systems.

Common Techniques:
a) Clustering
Groups similar data points together.
Examples: Customer segmentation, document grouping, image compression.
Algorithms: K-Means, Hierarchical Clustering, DBSCAN.
b) Association
Discovers rules that describe relationships between variables.
Examples: Market basket analysis ("customers who buy X also buy Y").
Algorithms: Apriori, FP-Growth.
c) Dimensionality Reduction
Reduces number of features while preserving important information.
Examples: Visualization, noise reduction.
Algorithms: PCA (Principal Component Analysis), t-SNE.
Summary for Unsupervised Learning:
Input (Features) → Model → Discover Patterns/Structures

Key Differences

Aspect	**Supervised Learning**	**Unsupervised Learning**
Data	Labeled (input-output pairs)	Unlabeled (only inputs)

Aspect	**Supervised Learning**	**Unsupervised Learning**
Goal	Predict output for new inputs	Discover hidden patterns or structures
Feedback	Direct feedback from known labels	No feedback; inferred structure
Complexity	Generally easier to train and evaluate	Often more subjective and challenging
Main Tasks	Classification & Regression	Clustering, Association, Dimensionality Reduction
Example Use Case	"Predict whether an email is spam (classification)."	"Group customers by purchasi behavior (clustering)."

When to Use Which?
Use Supervised Learning when you have labeled data and a clear prediction task (e.g., classifying images, forecasting values).
Use Unsupervised Learning when you want to explore data structure, find groupings, or reduce complexity without predefined labels.

5. Explain the difference between DM and ML.

6. Explain clustering, including common approaches such as K-means or Hierarchical clustering.

Clustering: Unsupervised Pattern Discovery

What is Clustering?

Clustering is an **unsupervised learning** technique that groups similar data points together based on their features, **without any prior knowledge of group labels**. The goal is to discover natural groupings (clusters) within the data where points in the same cluster are more similar to each other than to points in other clusters.

Key Idea: "Birds of a feather flock together."

Core Concepts in Clustering

1. Similarity/Distance Metrics

Clustering algorithms rely on measures of similarity or distance:

- **Euclidean Distance** (most common for K-means)
- **Manhattan Distance**
- **Cosine Similarity** (for text/directional data)
- **Jaccard Similarity** (for binary data)

2. Evaluation (No Ground Truth)

Since clusters aren't predefined, evaluation is challenging:

- **Internal Validation**: Silhouette score, Davies-Bouldin index

Silhouette Score - A Clustering Evaluation Metric

The **silhouette score** is a metric used to evaluate the quality of clusters created by clustering algorithms. It measures how similar an object is to its own cluster (cohesion) compared to other clusters (separation).

How It Works

For each data point i:

1. **a(i)**: Average distance between i and all other points in the same cluster
   - Measures cohesion (how close i is to its cluster members)
2. **b(i)**: Smallest average distance between i and points in any other cluster
   - Measures separation (how far i is from the nearest other cluster)
3. **Silhouette coefficient for point i**:

text

s(i) = (b(i) - a(i)) / max(a(i), b(i))

Interpretation

- **Range**: -1 to +1
- **+1**: Points are very well-clustered (tight clusters, far apart)
- **0**: Clusters are overlapping or indifferent
- **-1**: Points are likely assigned to wrong clusters

Formula

The overall silhouette score for the entire dataset is the average of all individual silhouette coefficients:

text

Silhouette Score = (1/n) * Σ s(i)

where n is the total number of points.

- **External Validation** (if labels exist): Adjusted Rand Index, Fowlkes-Mallows score
- **Visual Inspection**: Often crucial for interpretation

Common Clustering Approaches

1. K-Means Clustering

Most popular centroid-based algorithm

How It Works:

1. **Initialize**: Choose K cluster centers (centroids) randomly
2. **Assign**: Assign each point to nearest centroid
3. **Update**: Recalculate centroids as mean of assigned points
4. **Repeat**: Steps 2-3 until convergence (centroids stabilize)

Key Characteristics:

- **Requires specifying K** (number of clusters) in advance
- Creates **spherical, equally sized** clusters (due to Euclidean distance)
- Sensitive to **initial centroid placement** (often solved with K-means++)
- **Fast and scalable** to large datasets
- **Assumes clusters of similar density**

Choosing K:

- **Elbow Method**: Plot WCSS (Within-Cluster Sum of Squares) vs. K, look for "elbow"
- **Silhouette Analysis**: Measure how similar points are to their own cluster vs. others
- **Domain knowledge**: Based on practical requirements

Example Application:

python

```python
# Customer segmentation based on purchasing behavior

# K=3 might yield: "Budget Shoppers", "Regular Customers", "Premium Buyers"
```

Limitations:

- Poor performance with **non-spherical clusters**

- Sensitive to **outliers**
- Struggles with **clusters of different sizes/densities**
- **Hard assignment** only (each point belongs to exactly one cluster)

---

2. Hierarchical Clustering

Builds a tree-like structure (dendrogram) of clusters

Two Main Approaches:

a) Agglomerative (Bottom-Up) - Most Common

1. Start with each point as its own cluster
2. **Merge** two closest clusters repeatedly
3. Continue until one cluster remains or stopping criterion met

b) Divisive (Top-Down)

1. Start with all points in one cluster
2. **Split** clusters recursively
3. Continue until each point is its own cluster

Key Concepts:

- **Dendrogram**: Tree diagram showing cluster mergers/splits at different distances
- **Linkage Criteria** (how to measure distance between clusters):
  - **Single Linkage**: Minimum distance between points (can create "chain" clusters)
  - **Complete Linkage**: Maximum distance between points (creates compact clusters)

- **Average Linkage**: Average distance between points (balanced approach)
- **Ward's Method**: Minimizes variance within clusters (similar to K-means objective)

How to Use:

1. Build complete dendrogram
2. **Cut the dendrogram** at desired height to get specific number of clusters
3. No need to pre-specify K (can choose after seeing structure)

Advantages:

- Provides **hierarchical relationships** between clusters
- **Visual** representation via dendrogram aids interpretation
- No need to specify K upfront
- Can handle **non-spherical clusters** better than K-means

Disadvantages:

- **Computationally expensive** ($O(n^3)$ typically, $O(n^2)$ with optimization)
- Sensitive to **noise and outliers**
- **Once merged, clusters cannot be split** (agglomerative)
- Difficult with **very large datasets**

Example Application:

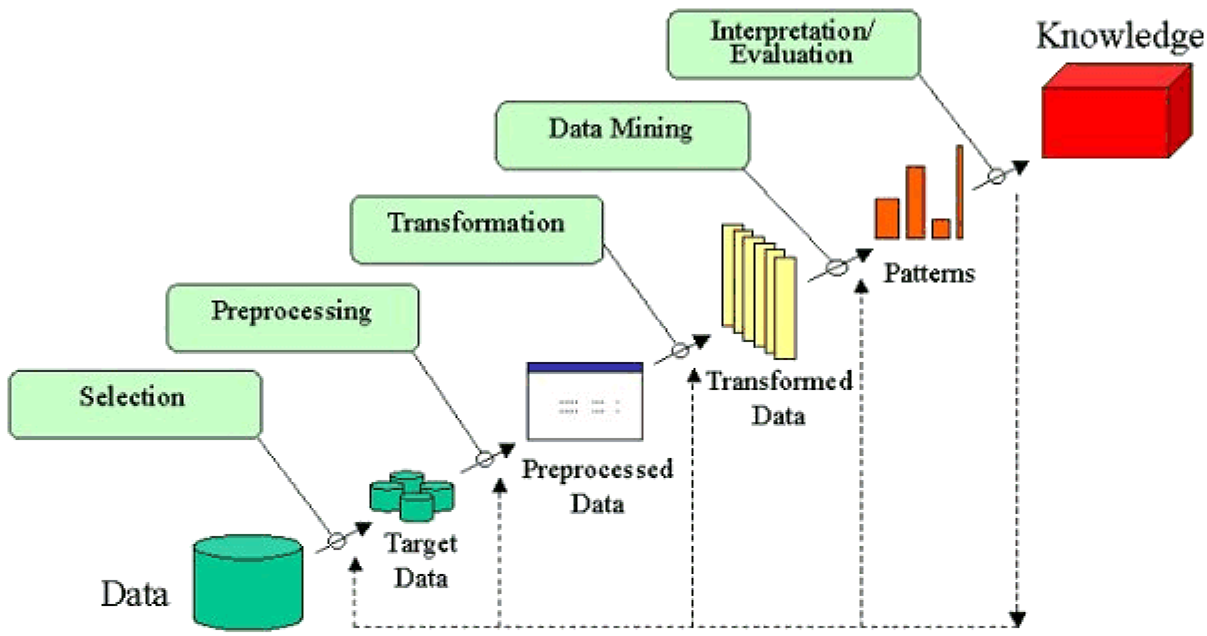
Comparison: K-Means vs. Hierarchical Clustering

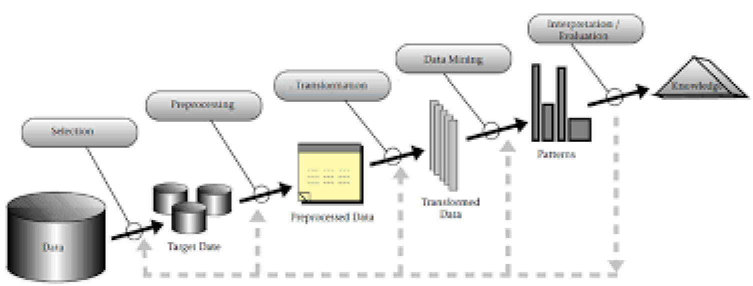
Aspect	K-Means	Hierarchical Agglomerative
Complexity	$O(n \times K \times I)$ - Linear scaling	$O(n^3)$ or $O(n^2)$ - Quadratic/cubic
K Specification	Must specify K beforehand	Choose K after analysis
Output Structure	Flat clusters	Hierarchical tree (dendrogram)
Cluster Shape	Prefers spherical, equal-sized	Adapts to various shapes
Best For	Large datasets, known K	Small/medium datasets, unknown K, need hierarchy
Result Stability	Multiple runs may give different results	Deterministic (same result each time)

Data Mining Life Cycles:

1. Explain Knowledge Discovery in Databases (KDD).

Knowledge Discovery in Databases (KDD)





Knowledge Discovery in Databases (KDD) is the overall process of discovering **useful, valid, and understandable patterns** from large volumes of data. It is not just data mining—data mining is only **one step** within the broader KDD process.

Definition

KDD is an **iterative, multi-step process** that transforms raw data into meaningful knowledge that can support decision-making.

Steps in the KDD Process

1. **Data Selection**
   Relevant data is selected from one or more databases, data warehouses, or data sources based on the problem domain.

2. **Data Preprocessing (Cleaning)**
   The selected data is cleaned to remove:

   - Noise
   - Missing values
   - Inconsistent data

3. **Data Transformation**
   Data is transformed into a suitable format for mining. This may include:

   - Normalization
   - Aggregation
   - Feature selection or reduction

4. **Data Mining**
   This is the core step where algorithms are applied to extract patterns, such as:

   - Classification
   - Clustering
   - Association rule mining
   - Regression

5. **Pattern Evaluation (Interpretation)**
   Discovered patterns are evaluated for usefulness and relevance. Redundant or uninteresting patterns are removed.

6. **Knowledge Presentation**
   The final knowledge is presented using visualization, reports, or dashboards so it can be easily understood by users.

Key Points

- KDD is **iterative**: results may lead back to earlier steps.
- Data mining is a **subset** of KDD.
- The goal is **actionable knowledge**, not just patterns.

Simple Flow (Text Diagram)

Data Sources

↓

Data Selection

↓

Preprocessing

↓

Transformation

↓

Data Mining

↓

Pattern Evaluation

↓

Knowledge

2. Outline the fundamental steps of KDD.

Knowledge Discovery in Databases (KDD)
KDD is the **overall process** of discovering useful, non-trivial, and actionable knowledge from large volumes of data. It is a **structured, iterative, and non-trivial** workflow that transforms raw data into meaningful patterns or knowledge.

Knowledge Discovery in Databases (KDD):
KDD is the overall process of discovering **useful, valid, and understandable knowledge** from large datasets. It is a multi-step methodology that transforms raw data into actionable insights.
Main stages of KDD:

1. **Data Selection** – Choosing relevant data from databases.
2. **Data Preprocessing** – Cleaning data (handling noise, missing values).
3. **Data Transformation** – Normalizing or reducing data for analysis.
4. **Data Mining** – Applying algorithms to extract patterns or models.
5. **Pattern Evaluation & Interpretation** – Identifying meaningful and useful knowledge.

👉 **Key point:** Data mining is **one step within KDD**, while KDD is the complete end-to-end knowledge discovery process.

Key Idea
KDD is **not just data mining**. Data mining is one essential step within the broader KDD process. KDD encompasses everything from data preparation to the final interpretation and deployment of discovered knowledge.

The KDD Process (Step-by-Step)

1. **Data Selection**
   - Identify and gather relevant data from various sources (databases, data warehouses, logs).
   - Focus on data subsets pertinent to the analysis goal.
2. **Data Preprocessing**
   - **Cleaning**: Handle missing values, remove noise, correct inconsistencies.
   - **Integration**: Combine data from multiple sources.
   - **Reduction**: Reduce data volume while preserving integrity (e.g., aggregation, dimensionality reduction).
3. **Data Transformation**
   - Convert data into forms suitable for mining.
   - Techniques: normalization, discretization, feature engineering, creating summary tables.
4. **Data Mining**
   - Apply algorithms to discover patterns.
   - Core tasks: classification, clustering, regression, association rule mining, anomaly detection.

5. **Interpretation/Evaluation**
    - Analyze mined patterns for validity, novelty, usefulness, and comprehensibility.
    - Visualize results.
    - Filter out irrelevant or redundant knowledge.
6. **Knowledge Deployment**
    - Integrate the discovered knowledge into decision-making systems, reports, or operational processes.
    - Example: deploying a fraud detection model or a recommendation engine.

KDD vs. Data Mining

- **KDD** is the **entire process** (from raw data to actionable knowledge).
- **Data Mining** is the **specific step** (applying algorithms to find patterns) within KDD.

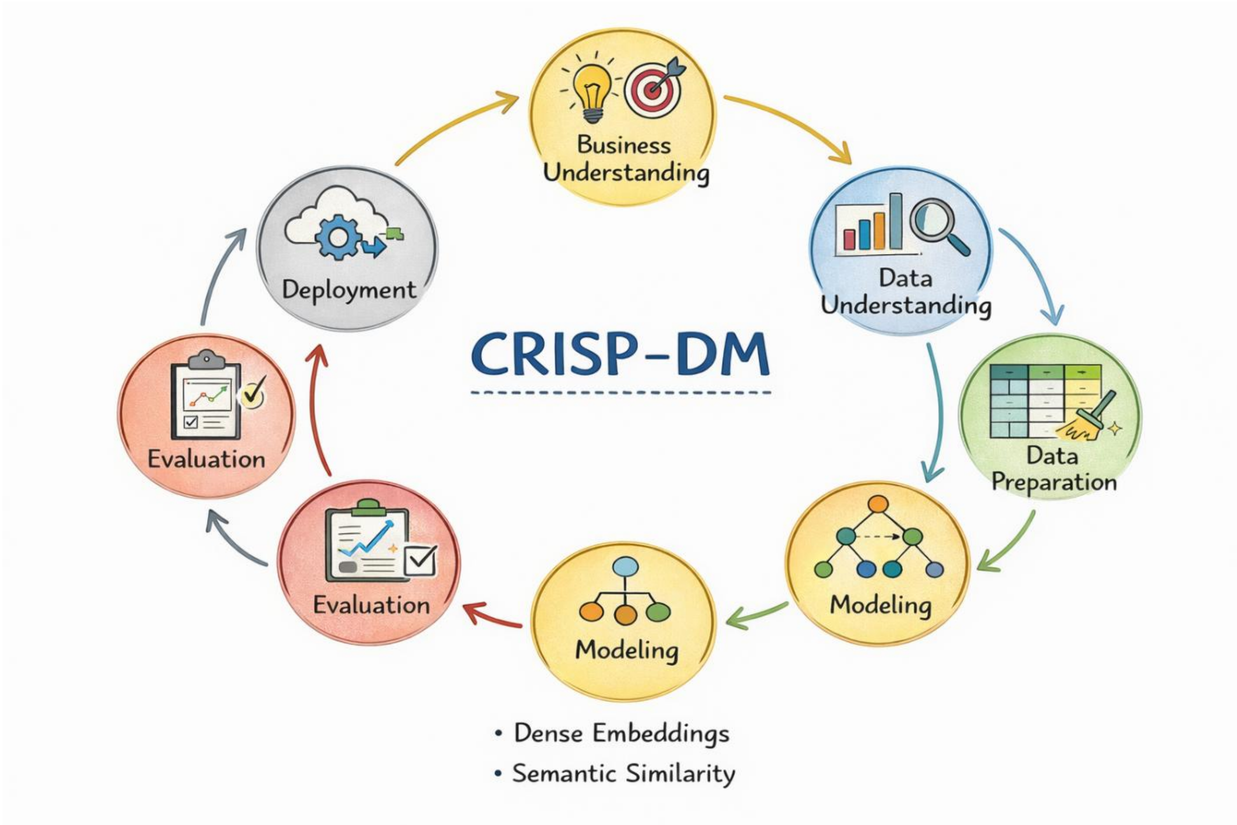
Real-World KDD Example: Retail Customer Segmentation

1. **Selection**: Extract customer transaction records and demographics.
2. **Preprocessing**: Clean missing values, integrate loyalty card data.
3. **Transformation**: Create features like “average purchase value” or “frequency.”
4. **Data Mining**: Apply clustering (e.g., k-means) to group similar customers.
5. **Interpretation**: Analyze clusters (e.g., “high-value frequent buyers”).
6. **Deployment**: Use segments for targeted email marketing campaigns.

Importance of KDD

- Provides a **systematic framework** to ensure discoveries are reliable and actionable.
- Emphasizes that **data preparation and interpretation** are as crucial as the mining algorithm itself.
- Iterative nature allows refinement based on feedback and new data.

3. Introduce CRISP-DM (Cross-Industry Standard Process for Data Mining) and



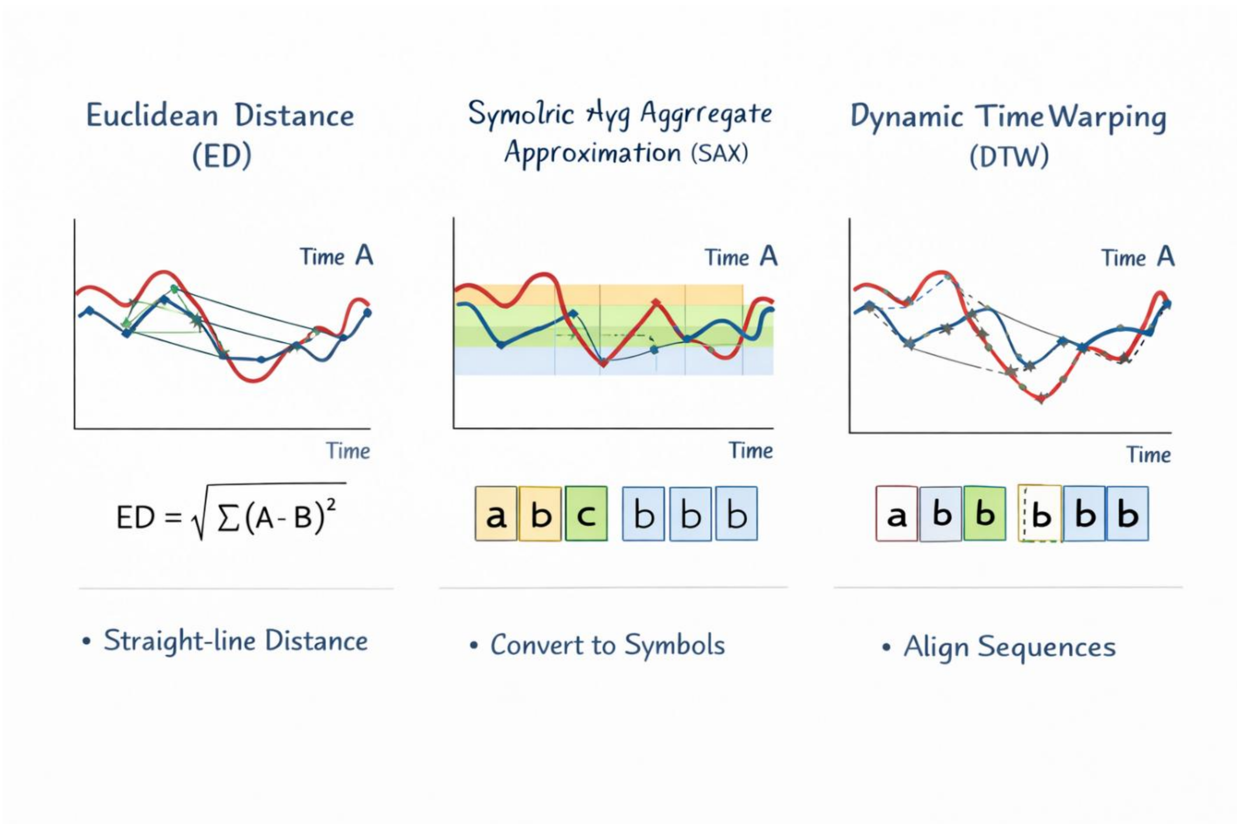
elaborate on its primary phases.

Data Preprocessing:

1. Define data preprocessing.
2. Discuss techniques for handling missing values.
3. Explore data discretization, cleaning, integration, and transformation methods.
4. Unpack the concept of feature selection, covering methods like Filters, Wrappers, PCA, and DL. Explain when it is best to use each approach.
5. Explain the role of entropy and Information Gain in Decision Trees.

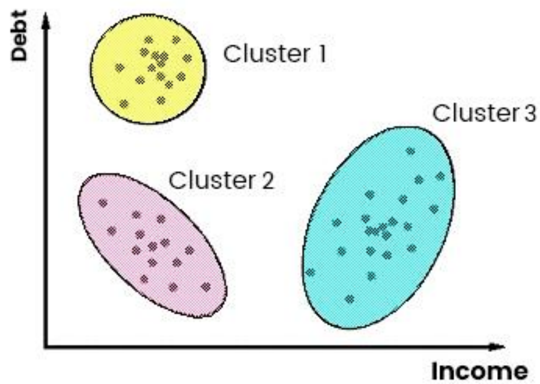
Time Series:

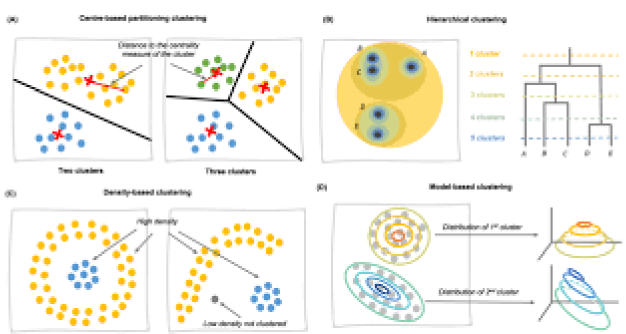
1. Define time series (TS).
2. Examine similarity measures.
3. Explore feature extraction in TS, emphasizing the significance of windows.
4. Introduce Dynamic Time Warping (DTW) and Symbolic Aggregate Approximation (SAX).

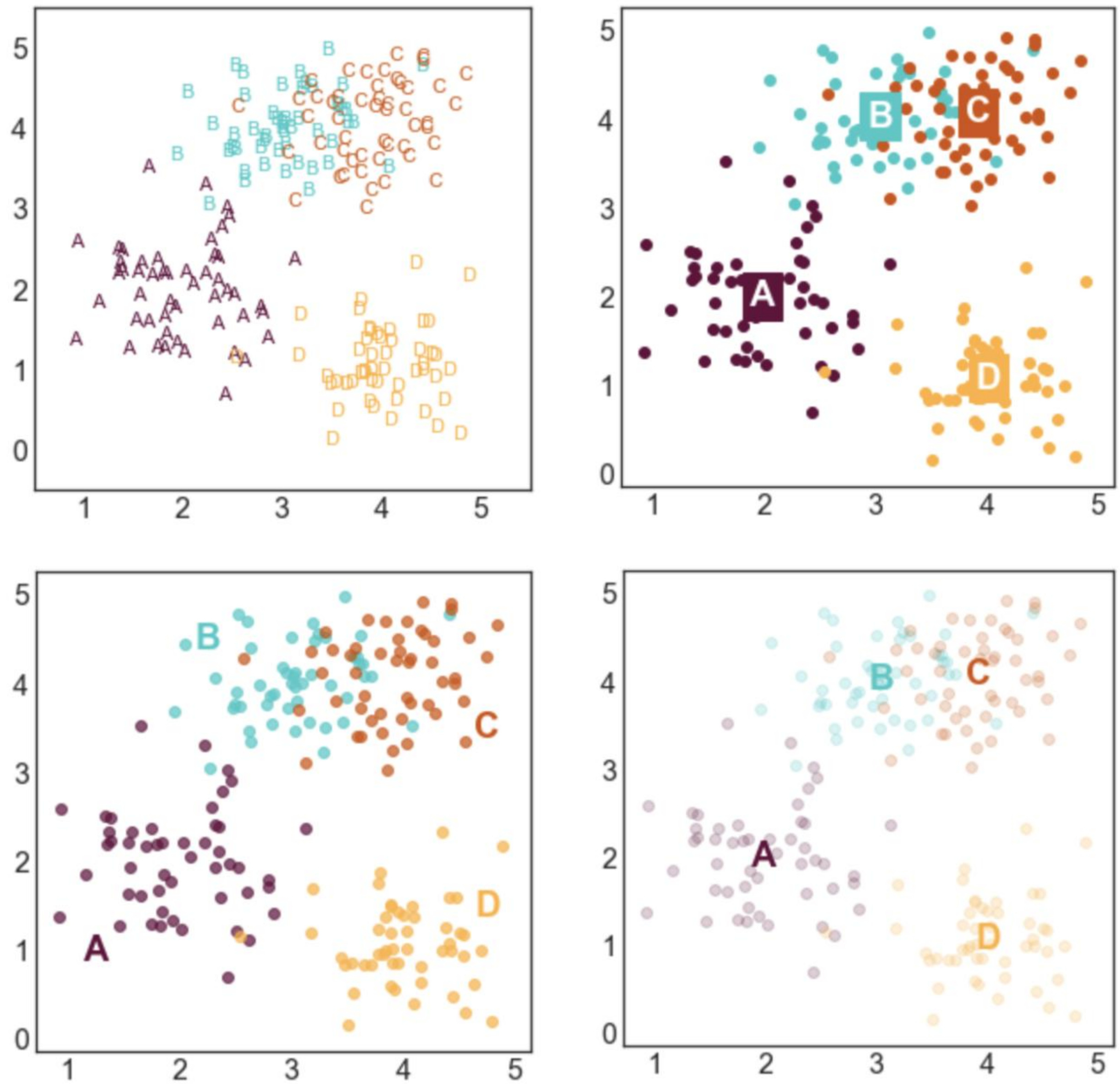


Clustering:

1. Clustering and its purpose, including hierarchical clustering with dendrograms.
   Clustering and Its Purpose







Clustering is an **unsupervised learning technique** in data mining that groups a set of data objects into clusters so that:

- Objects **within the same cluster** are highly similar
- Objects **in different clusters** are highly dissimilar

Unlike classification, clustering works **without predefined class labels**.

Purpose of Clustering

Clustering is used to:

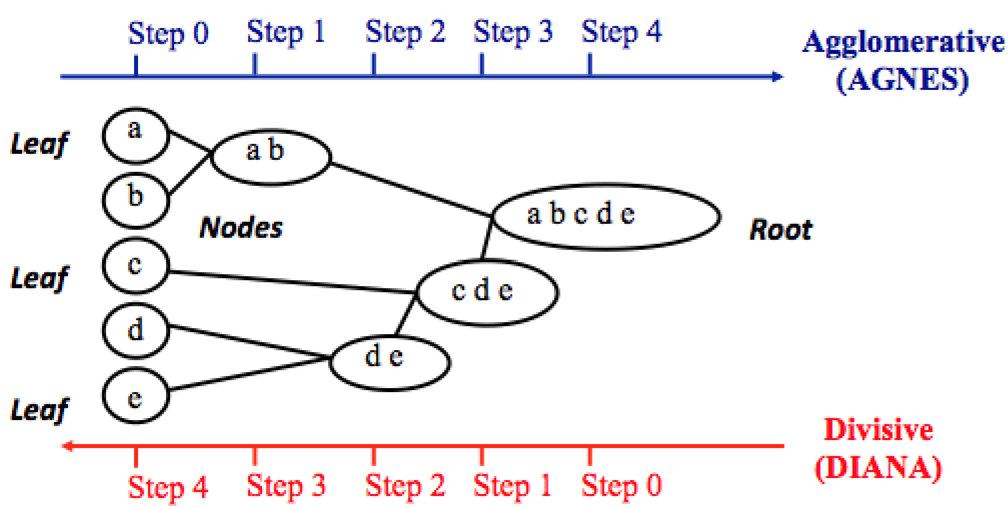
- Discover **hidden patterns** in data
- Group similar customers, documents, or images
- Perform **data summarization**

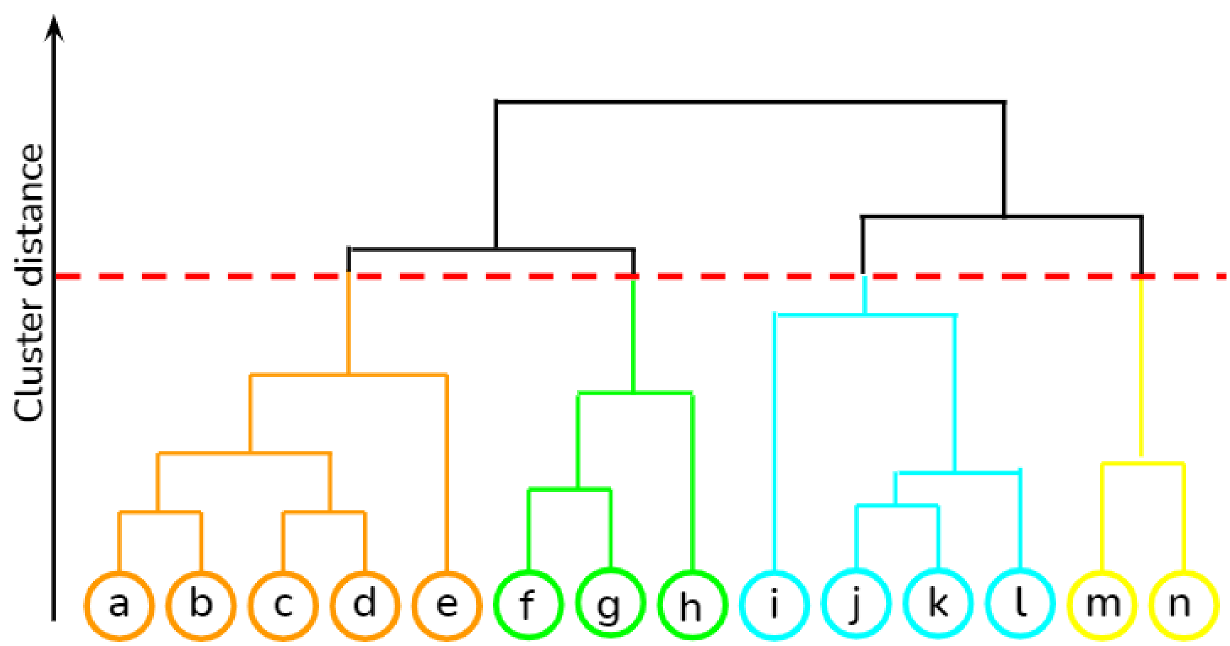
- Detect **outliers or anomalies**
- Support decision-making and business intelligence

Applications:

- Market segmentation
- Image segmentation
- Social network analysis
- Bioinformatics
- Document and web mining

Hierarchical Clustering





Hierarchical clustering builds a hierarchy of clusters represented as a **tree structure called a dendrogram**.

Types of Hierarchical Clustering

1. **Agglomerative (Bottom-Up)**
   - Each data object starts as a single cluster
   - Closest clusters are repeatedly merged
   - Continues until one large cluster remains
2. **Divisive (Top-Down)**
   - All data objects start in one cluster
   - The cluster is repeatedly split into smaller clusters

Steps in Agglomerative Hierarchical Clustering

1. Treat each data point as a separate cluster
2. Compute distance between all clusters
3. Merge the two closest clusters
4. Update distance matrix
5. Repeat until stopping criterion is met

Dendrogram

A **dendrogram** is a tree-like diagram that:

- Shows the **order of cluster merging**
- Displays the **distance (similarity level)** at which clusters are joined
- Helps decide the **number of clusters** by cutting the tree at a chosen height

---

Dendrogram Diagram (Text Representation)

Distance

```
 |
 |       --------- Cluster A
 |       |
 |   ----|     ---- Cluster B
 | |     |
 |--|    --------- Cluster C
 |
 +------------------------------ Objects
      1    2    3    4    5
```

Cutting the dendrogram horizontally at a certain distance gives the desired number of clusters.

Advantages of Hierarchical Clustering

- No need to predefine number of clusters
- Easy to interpret using dendrograms
- Works well for small datasets

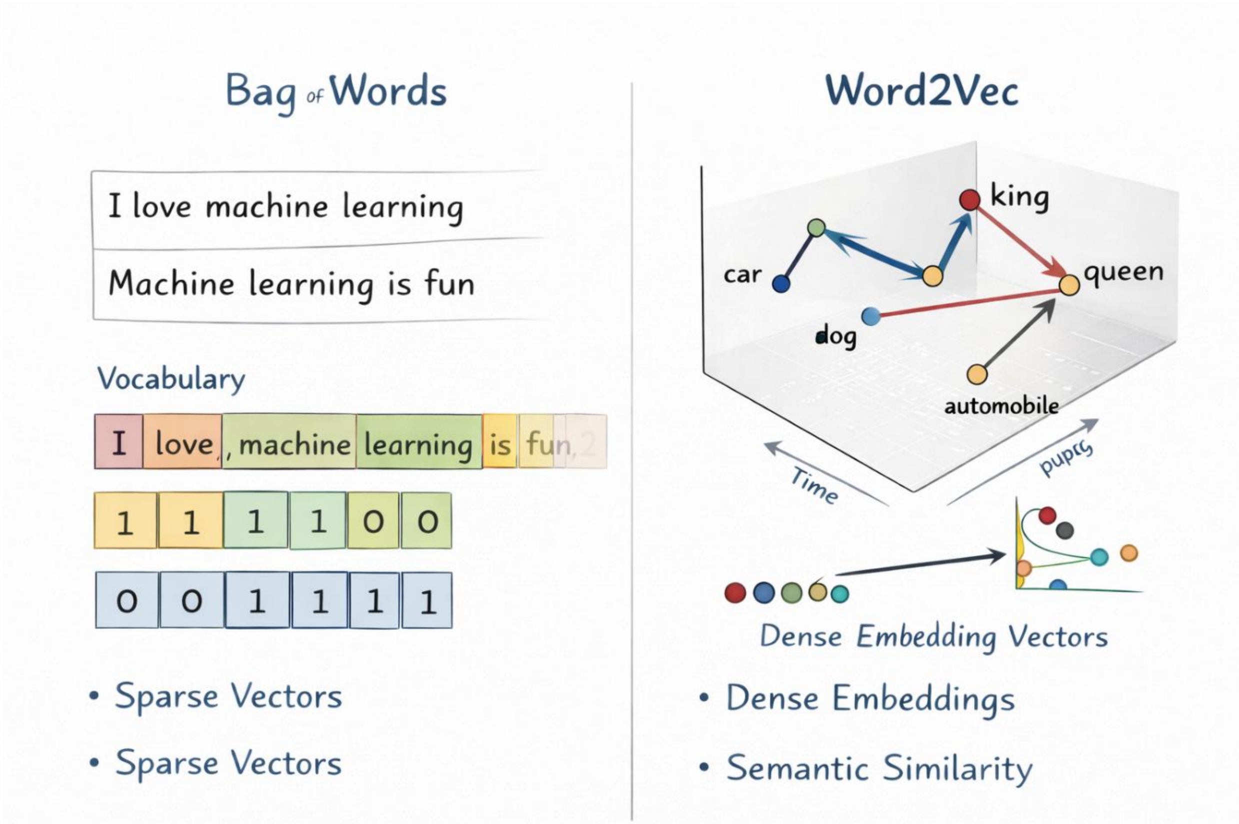
Limitations

- Computationally expensive for large datasets
- Once a merge or split is made, it **cannot be undone**

2. Types of clustering, such as time series k-means, partitional k-means, and hierarchical clustering.
3. K-means algorithm, covering its objective and iterative steps.

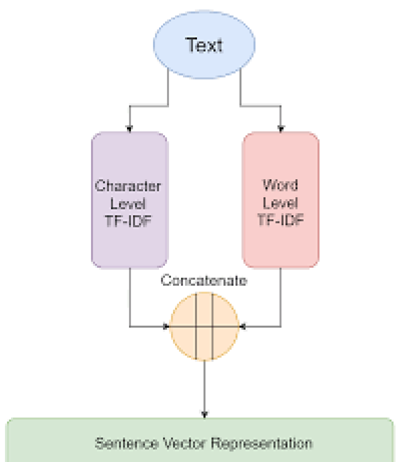
Text Mining:

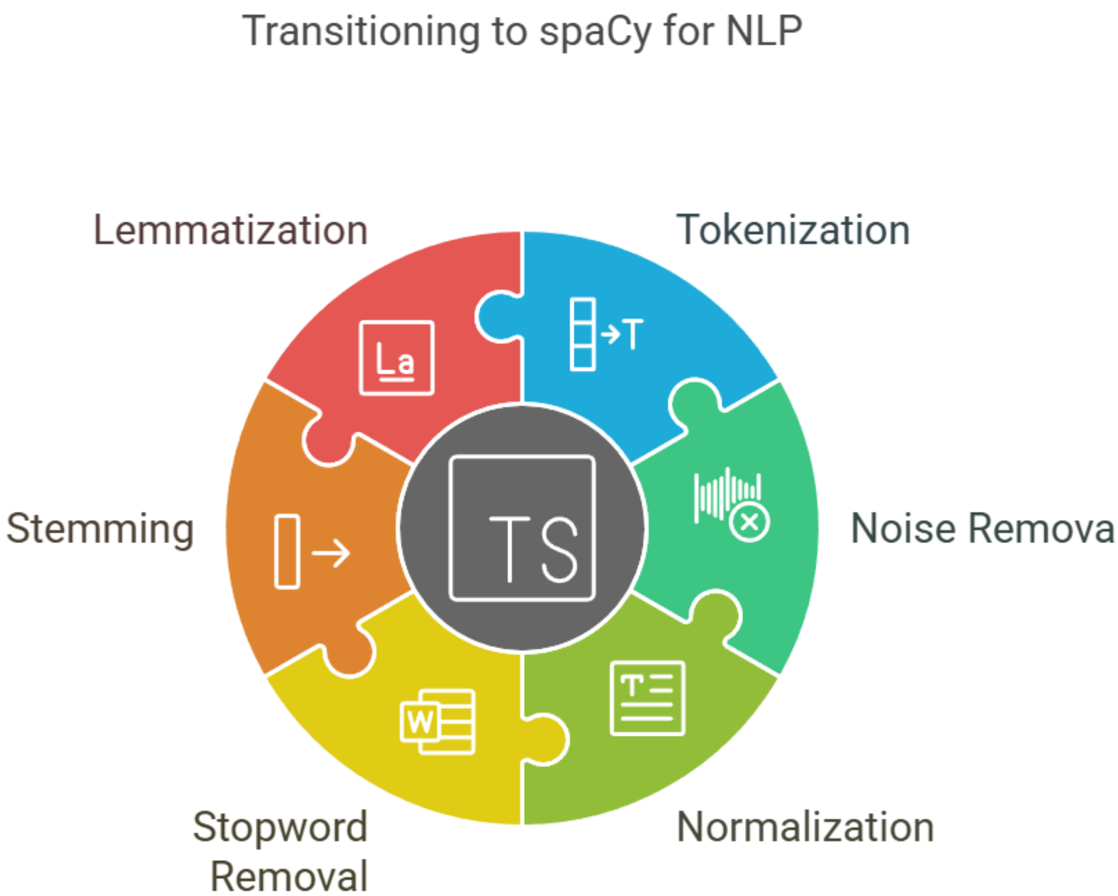
1. Please provide definitions and applications for text mining and how they relate to NLP.



2. Explain the issues that supervised learning has with text.
3. Provide definitions for Bag of words, TFIDF, word embedding, transfer learning, stop words, lemmatisation, stemming, tokenisation, and pos tagging.
   Definitions of Common NLP Terms (with diagrams)


engati
Simply Intelligent
What is
Bag of words?
document id
doc_id 456


Text
Character
Level
TF-IDF
Word
Level
TF-IDF
Concatenate
Sentence Vector Representation



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1. Bag of Words (BoW)

Bag of Words is a text representation technique where a document is represented by the **frequency of words**, ignoring grammar and word order.

- Each document → vector of word counts
- Simple and widely used
- Does not capture meaning or context

Example:

Text: *"I like NLP and I like data"*

BoW: {I:2, like:2, NLP:1, and:1, data:1}

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2. TF-IDF (Term Frequency–Inverse Document Frequency)

TF-IDF measures how **important a word is** in a document relative to a collection of documents.

- **TF**: Frequency of term in a document
- **IDF**: Penalizes common words across documents

$$TF\text{-}IDF = TF \times IDF$$

Helps reduce the effect of common but less informative words.

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3. Word Embedding

Word embedding represents words as **dense numerical vectors** in a continuous vector space where semantic similarity is preserved.

- Similar words → vectors close to each other
- Captures context and meaning
- Examples: Word2Vec, GloVe, FastText

Example:

king − man + woman ≈ queen

---

4. Transfer Learning

Transfer learning is a technique where a **pre-trained model** is reused for a new but related task.

- Saves time and computational cost
- Improves performance with less data
- Common in NLP (e.g., BERT, GPT)

Example:

Using a pre-trained language model for sentiment analysis.

---

5. Stop Words

Stop words are commonly used words that usually carry **little semantic meaning**.

- Examples: *is, the, am, and, of*
- Removed to reduce noise and dimensionality

---

6. Lemmatization

Lemmatization reduces words to their **base (dictionary) form**, considering grammar and context.

- Accurate but slower than stemming

Examples:

- running → run
- better → good

---

7. Stemming

Stemming reduces words to their **root form** by removing suffixes.

- Faster but less accurate
- May produce non-meaningful words

Examples:

- running → run
- studies → studi

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8. Tokenization

Tokenization is the process of breaking text into **smaller units** called tokens.

- Tokens can be words, subwords, or sentences

Example:

Text: *“I love NLP”*

Tokens: [I, love, NLP]

---

9. POS Tagging (Part-of-Speech Tagging)

POS tagging assigns **grammatical labels** to words in a sentence.

- Identifies nouns, verbs, adjectives, etc.
- Helps in syntactic and semantic analysis

Example:

“NLP is fun”

- NLP → Noun
- is → Verb
- fun → Adjective

Overall NLP Processing Flow (Diagram – Text)

Raw Text
↓
Tokenization
↓
Stop Word Removal
↓
Stemming / Lemmatization
↓
POS Tagging
↓
Feature Extraction (BoW / TF-IDF / Embeddings)

Explainability:

1. Differentiate between Explainability and Interpretability.

   Interpretability vs. Explainability :

- **Interpretability:** The model is *inherently understandable*—humans can directly see how inputs map to outputs (e.g., linear regression, decision trees).
- **Explainability:** *Post-hoc methods* are used to explain a model's behavior after the fact, often for complex or black-box models (e.g., SHAP, LIME for neural networks).

In short:
Interpretability is built in; explainability is added on.

2. Relationship of Interpretability with the performance of models.

Relationship Between Interpretability & Model Performance
The Core Trade-off Diagram
text

```
          MODEL COMPLEXITY vs PERFORMANCE vs INTERPRETABILITY

 Performance ↑
   (Accuracy, F1, AUC)  |
                   |        Deep Neural Networks
                   |             ●
                   |
       Random      |              ● Ensemble Methods
       Forests ●   |
                   |         ●
     Decision ●    |           ● SVM (RBF kernel)
     Trees         |
                   |    ●
       Logistic    |     ●
       Regression ●|       ●
       Linear      |●
       Regression  |─────────────────────┐
                   |       Interpretability
                   |       (High → Low)
                   └────────────────────────────────────→
```

Key Relationships
1. Inverse Relationship (Typically)

text
Complexity ⬆ → Performance ⬆ → Interpretability ⬇
Simple Model → Interpretability ⬆ → Performance ⬇ (for complex patterns)

2. Performance-Interpretability Matrix

text
```
               HIGH PERFORMANCE
                     │
             ┌───────────┼───────────┐
             │     │     │
   LOW       │ Modern │ State- │
INTERPRETABILITY │ DL/  │ of-the-│ HIGH
             │ XGBoost│ Art  │ INTERPRETABILITY
             │     │     │
             │───────────┼───────────│
             │     │     │
             │ Trade- │ Linear │
             │ off  │ Models │
   LOW       │ Zone │ Rule-  │
 PERFORMANCE │      │ Based  │
             └───────────┴───────────┘
```

Why the Trade-off Exists

Black Box Models (High Performance)

- **Examples**: Deep Neural Networks, Gradient Boosting, Complex Ensembles
- **Performance Advantages**:
  - Capture complex non-linear patterns
  - Automatic feature engineering
  - Handle high-dimensional data well
- **Interpretability Costs**:
  - Decisions are opaque
  - Hard to explain "why"
  - Feature importance is indirect

Glass Box Models (High Interpretability)

- **Examples**: Linear Regression, Decision Trees (small), Rule-based systems
- **Interpretability Advantages**:
  - Clear decision rules
  - Transparent feature importance
  - Easy to audit and validate
- **Performance Limitations**:
  - Struggle with complex patterns
  - Require manual feature engineering
  - May oversimplify relationships

3. Define and exemplify accountability, trustworthiness, causality, confidence, fairness, and ethical decision-making.

Accountability, Trustworthiness, Causality, Confidence, Fairness & Ethical Decision-Making in AI

1. Accountability

Definition: The obligation to explain, justify, and take responsibility for AI system outcomes and impacts.

Key Components:

- **Answerability**: Ability to explain decisions
- **Responsibility**: Clear ownership of outcomes
- **Remediation**: Processes to address harms
- **Auditability**: Transparent record-keeping

Real-world Example:

- **Tesla Autopilot** - Must maintain data logs showing system state before accidents
- **Credit scoring algorithms** - Legally required to explain denial reasons under Equal Credit Opportunity Act
- **Healthcare AI** - FDA requires validation, monitoring, and adverse event reporting

2. Trustworthiness

Definition: The degree to which AI systems are reliable, safe, transparent, and worthy of user confidence.

Four Pillars:

text

```
Trustworthiness =
├── Reliability (Consistent performance under varying conditions)
├── Security (Resistance to attacks/manipulation)
├── Privacy (Data protection)
└── Transparency (Understandable operations)
```

- **IBM Watson Health** - Extensive validation against clinical trials
- **Google Search** - Consistent results across billions of queries
- **Signal/Telegram** - End-to-end encryption for privacy

3. Causality

Definition: Understanding cause-effect relationships rather than just correlations in AI predictions.

Key Distinction:

- **Correlation**: "When ice cream sales increase, drowning deaths increase" (Both caused by summer)
-
- **Real-world Example**:
- **Drug discovery** - Distinguishing causal therapeutic effects from correlated biomarkers
- **Economic policy** - Understanding if minimum wage increases cause unemployment
- **Marketing analytics** - Determining if ads actually cause purchases vs. brand awareness

4. Confidence

Definition: The AI system's self-assessment of prediction reliability, often expressed as probability scores or uncertainty measures.

Types of Confidence:

1. **Aleatoric uncertainty** - Inherent data noise
2. **Epistemic uncertainty** - Model ignorance
3. **Distributional uncertainty** - Out-of-distribution data

Real-world Example:

- **Weather prediction** - "70% chance of rain" with confidence intervals
- **Self-driving cars** - Slowing down when confidence in object recognition drops
- **Spam filters** - Moving emails to "spam" only when confidence > 99%

5. Fairness

Definition: Ensuring AI systems don't create or reinforce unfair bias, discrimination, or disadvantage against individuals or groups.

Fairness Metrics:

text

Demographic Parity: $P(\hat{Y}=1|A=\text{male}) = P(\hat{Y}=1|A=\text{female})$

Equal Opportunity: $P(\hat{Y}=1|Y=1,A=\text{male}) = P(\hat{Y}=1|Y=1,A=\text{female})$

Equalized Odds: Both TPR and TPR equal across groups

Real-world Example:

- **Amazon recruiting tool** - Discarded after showing bias against women
- **COMPAS recidivism algorithm** - Controversy over racial bias in risk scores
- **Facebook ad delivery** - Settled lawsuits over discriminatory housing ads

6. Ethical Decision-Making

Definition: The process by which AI systems make choices aligned with moral principles, values, and human ethics.

Ethical Frameworks:

1. **Utilitarian**: Maximize overall good
2. **Deontological**: Follow moral rules/duties
3. **Virtue ethics**: Cultivate moral character
4. **Care ethics**: Prioritize relationships and care

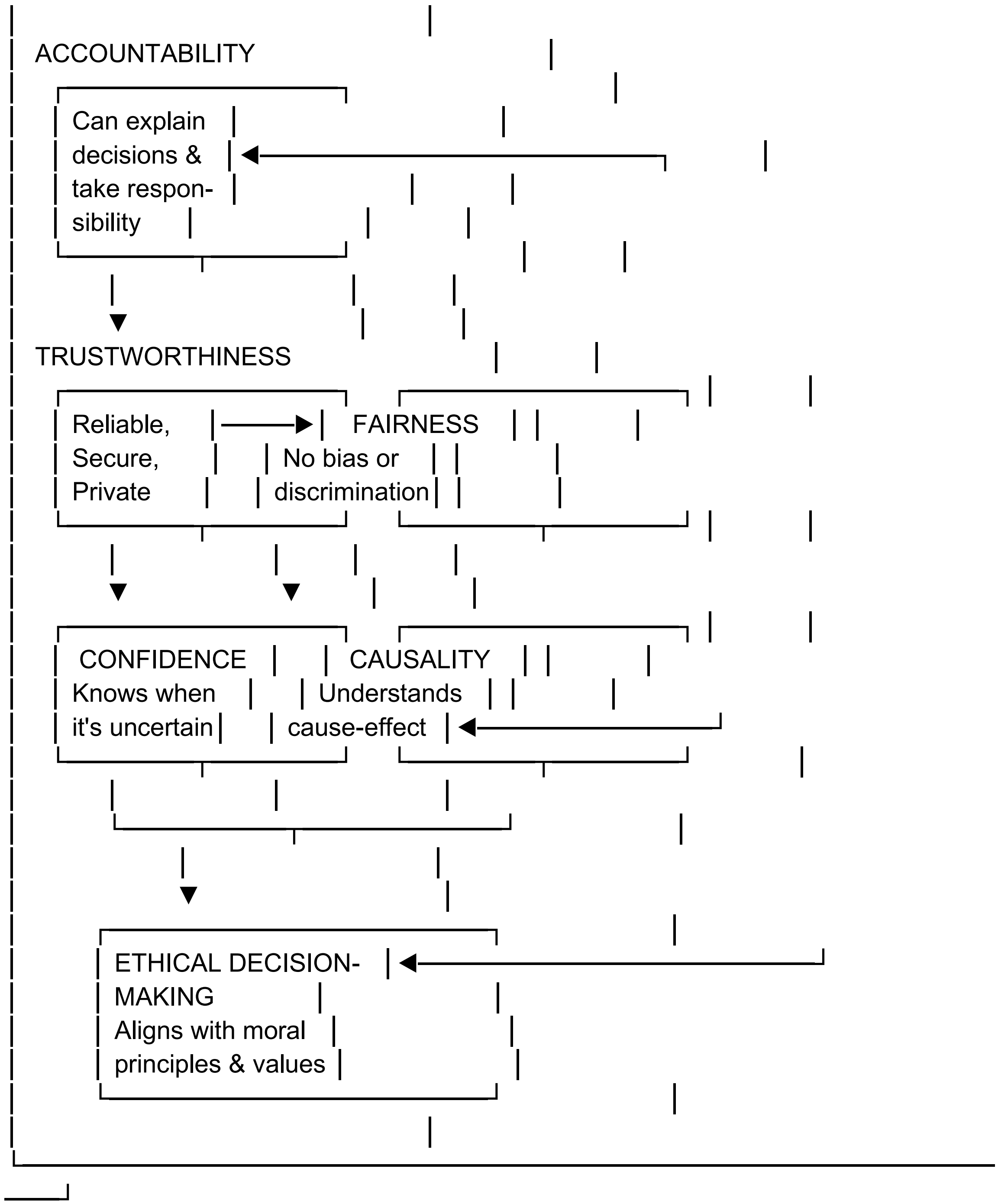
Real-world Ethical Frameworks:

- **Asilomar AI Principles** - 23 guidelines for beneficial AI
- **EU Ethics Guidelines for Trustworthy AI** - 7 requirements
- **Google's AI Principles** - Prohibit weapons, require social benefit
- **IEEE Ethically Aligned Design** - Human rights focused approach

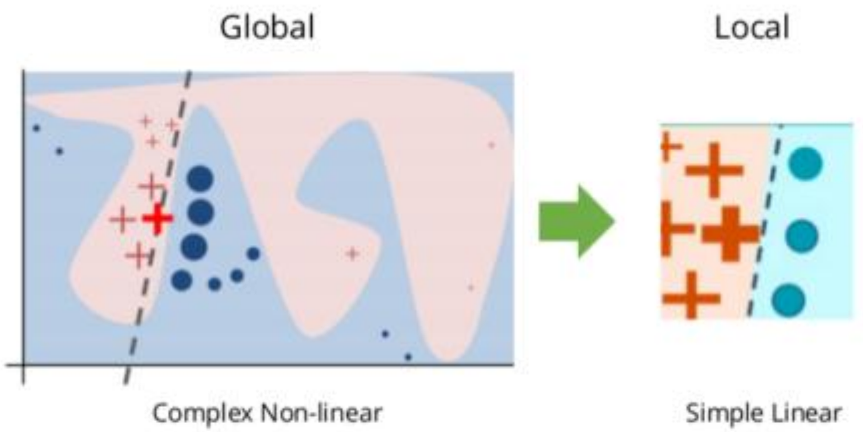
Interconnections Diagram

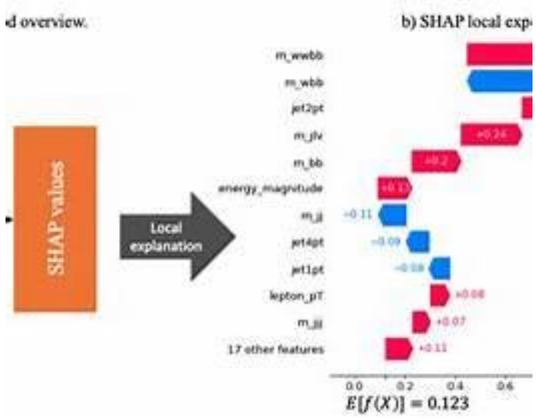
text

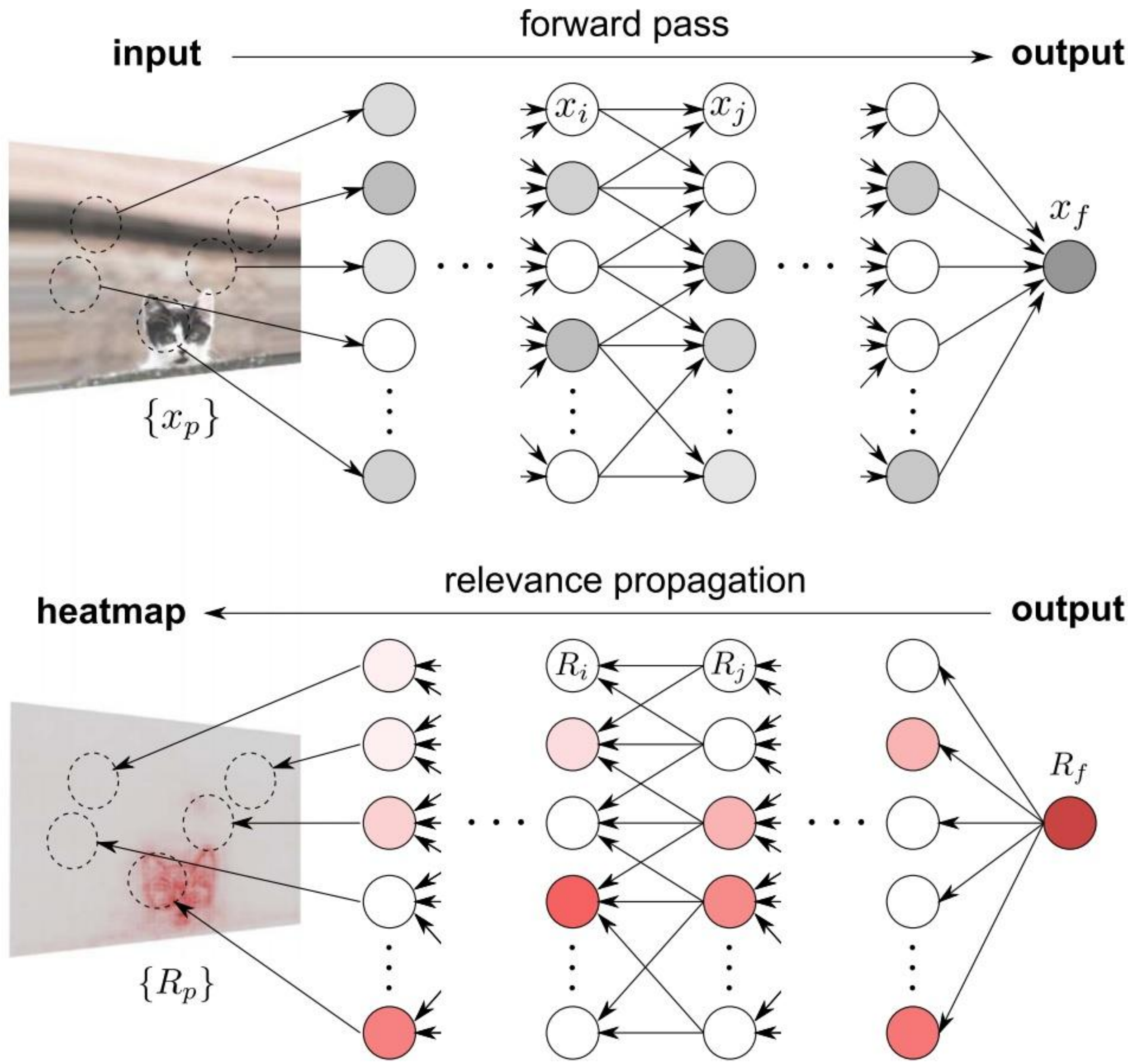
```
┌──────────────────────────────────────────────────────────
────┐
│           ETHICAL AI ECOSYSTEM                  │
├──────────────────────────────────────────────────────────
────┤
```



4. Provide examples of explainability methods, including LIME, SHAP, and LRP.
   Explainability Methods (with Examples)


Global
Local
Complex Non-linear
Simple Linear


d overview.
b) SHAP local exp
SHAP values
Local explanation
m_wwbb
m_wbb
jet2pt
m_jlv
m_bb
energy_magnitude
m_jj
jet4pt
jet1pt
lepton_pT
m_jjj
17 other features
+0.24
+0.2
-0.11
+0.08
+0.07
+0.11
0.0
0.2
0.4
0.6
E[f(X)] = 0.123



Explainability methods help us understand *why* a machine-learning model makes a particular prediction, especially for complex or black-box models.

---

1. LIME (Local Interpretable Model-agnostic Explanations)
Definition:
LIME explains an individual prediction by **approximating the complex model locally** with a simple, interpretable model (like linear regression).
How it works:

- Selects one prediction
- Perturbs the input data around it
- Learns a simple model to explain that local behavior

Example:
For a **spam classifier**, LIME may show that words like *"free"* and *"win"* contributed most to classifying an email as spam.

---

2. SHAP (SHapley Additive exPlanations)
Definition:
SHAP explains predictions using **Shapley values** from game theory, fairly distributing a prediction's contribution among features.
How it works:

- Considers all possible feature combinations
- Assigns each feature a contribution value

Example:
In **credit scoring**, SHAP might show:

- Income: +0.25 (increases approval)
- High debt: −0.40 (decreases approval)

SHAP can explain both **individual predictions** and **global model behavior**.

---

3. LRP (Layer-wise Relevance Propagation)
Definition:
LRP explains predictions in **neural networks** by propagating the prediction score **backward through the layers**, assigning relevance to each input feature.
How it works:

- Starts from output layer
- Distributes relevance back to inputs layer by layer

Example:
In **image classification**, LRP highlights the pixels that most influenced the decision (e.g., parts of an image responsible for identifying a cat).

Summary Table

Method	Model Type	Explanation Scope	Example Use
LIME	Model-agnostic	Local	Text, tabular
SHAP	Model-agnostic	Local & Global	Finance, healthcare
LRP	Neural networks	Local	Image, deep learning

5. Contrast model-agnostic and model-specific approaches.

6. Distinguish between Local and Global explainability methods.

7. Discuss Pre-model, In-model, and Post-model Strategies.